

NITHIN GOWDA K N

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🌐 <https://nithingowda-portfolio>

Summary

Machine Learning Engineer with experience designing and deploying AI systems for predictive modeling and automation. Proficient in developing ML pipelines, optimizing deep learning models, and working with generative AI technologies, including LLMs, RAG and Agents. Demonstrated expertise in model training, evaluation, API development, and full-stack ML integration. Skilled in statistical analysis, model fine-tuning, and software engineering best practices.

Experience

Datafoundry

Nov 2024 – Present

Machine Learning engineer

Bangalore, Karnataka

- Strong command of Python or Javascript, including libraries/tools for data science and machine learning.
- Proficiency with TensorFlow, PyTorch, or Scikit-learn for building and deploying models.
- Built REST APIs, microservices and React.js-based UIs to integrate AI/ML models into business solutions.
- Engineered production RAG pipelines using FAISS, Weaviate, and HuggingFace embeddings for intelligent retrieval.
- Built LLM-powered assistant agents using Ollama, LLama3, Mistral, Gemini, and integrated ReAct reasoning flows.
- Applied LangChain, LangGraph, and HuggingFace to coordinate multi-agent workflows with modular backends.
- Developed an OCR pipeline using Tesseract and EasyOCR to retain text orientation and structure, significantly improving downstream LLM performance in document understanding tasks.
- Developed crawlers, scrapers, and data pipelines to extract structured insights from medical/legal content.
- Worked on data preprocessing, feature engineering, model training, validation and deployment.
- Collaborate with Data Engineers, Data Scientists, and other stakeholders to implement end-to-end AI/ML solutions.
- Experience in a collaborative development environment, using version control tools like Git.
- Utilize deep learning, NLP, computer vision, and time-series analysis to develop innovative solutions.
- Maintained scalable Conversational AI services using Rasa, Dockerized microservices, and Git-based deployments.
- Strong problem-solving abilities and analytical skills to identify and address technical challenges.

Dr Ali's Denta Care

Dec 2023 – Feb 2024

Freelance

Bangalore, Karnataka

- Collected and labeled dental X-ray datasets; performed augmentation and preprocessing using OpenCV.
- Fine-tuned CNN-based object detectors (YOLOv5, Detectron2) to localize cavities with over 91% accuracy on test sets.
- Designed Python-PyTorch pipelines for image classification, validation, and export to lightweight mobile formats.

Projects

LLM-Based Chatbot with Memory and Web Search Integration | *Python, React js, LangGraph*

May 2025

- Built a chatbot using React and Gemini LLM for real-time, dynamic frontend-backend interaction.
- Used LangGraph to implement memory-enabled chatbot with multi-turn conversation support.
- Integrated a web search tool for fallback answers when chatbot lacks direct knowledge of queries.

ReAct-RAG based Chatbot for Indian Legal Documents | *Python, RAG, FAISS*

June 2025

- Built full-stack legal chatbot with React frontend and LangGraph backend for agentic flow handling.
- Designed ReAct-RAG agent using Gemini, FAISS, and Sentence-Transformers for semantic search.
- Enabled retrieval over Indian Constitution and IPC PDFs using vector-based document search.

Multi-Agent IPL Chatbot with Gemini and Text-to-SQL | *Python, Javascript, Gemini*

May 2025

- Developed a multi-agent system using Gemini for IPL query handling and orchestration logic.
- One agent orchestrates tasks, while another converts natural language queries into SQL format.
- Integrated web search tool to handle unknown queries beyond the agents' existing knowledge.

Movie Recommender System | *Python, Streamlit*

sep 2024

- Built a Streamlit-based movie recommender using collaborative filtering and cosine similarity.
- Recommended similar movies based on user query for personalized viewing suggestions.

Technical Skills

Languages: Python, C++, HTML/CSS, JavaScript, SQL.

ML/Deep Learning: CNN, YOLO, Scikit-learn, pytorch, TensorFlow, NLP, Computer Vision.

Gen AI: Ollama, OpenAI, LangChain, Multimodal LLMs, Prompt Engineering, RAG, AI Agents, Conversational AI.

Other: Micro Services, Git, Docker, REST APIs, Celery, Redis DB, Data Structures.

Education

University of Mysore

Jan. 2022 – Sep 2023

Masters of Science in Computer Science

Mysore, Karnataka